

Research Statement

CURRENT RESEARCH

My research in Human–Computer Interaction (HCI) focuses on reimagining the human body as an intelligent interface for virtual and mixed reality. While avatars provide a recognizable metaphor for embodiment, they often reduce interaction to representational surrogates. My work challenges this avatar-centric paradigm by developing novel bodily interfaces that extend action, perception, and control in immersive systems. By designing and evaluating these techniques, I aim to establish a new interaction paradigm where the body is not merely represented but actively augmented to enable efficient, precise, and sustainable forms of interaction. This research contributes to HCI by broadening the design space of embodied interfaces, to cognitive science by showing how bodily awareness shapes perception and memory, and to industry by offering principles for practical XR applications.

The first strand of my work expands bodily reach through a distributed control paradigm that reimagines the hand not as a continuous avatar but as a swarm of controllable points. In this “Swarm Hand” approach, the hand becomes a dynamic field capable of enveloping and manipulating multiple entities at once. By adjusting the distribution of swarm particles, users can gather, rotate, or resize groups of objects in a single gesture, performing tasks more efficiently than with either controller rays or conventional virtual hands. Controlled experiments reveal significant time savings in multi-target selection and resizing, while qualitative feedback highlights the technique’s flexibility, immersion, and sense of agency [6]. This demonstrates that bodily interaction can be reconceptualised as distributed control rather than unitary manipulation, introducing a new framework for multi-object interaction.

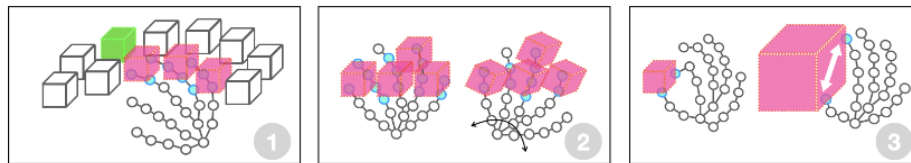


Figure 1: The Swarm Hand [5, 6] reconceptualizes the hand as a dynamic field of particles, enabling users to envelop, gather, and manipulate multiple virtual objects simultaneously. Compared to conventional avatar hands or ray-casting controllers, this distributed approach supports faster multi-object manipulation while enhancing flexibility and immersion.

A second strand embeds interaction directly on the body surface through the design of on-body menus. Digital functions are anchored to bodily landmarks such as arms, torso, or legs, transforming the body into a mnemonic canvas for interaction. I studied how different perspectives—first-person, third-person, and mirrored—shape the creation and recall of these menus. Experiments show that mirrored perspectives support faster creation and more accurate recall, while first-person views promote a stronger sense of involvement. Beyond performance, consistent patterns of icon placement emerged: participants placed social applications on the forearms, productivity tools on the torso, and miscellaneous functions on the lower body. These findings demonstrate that on-body menus exploit embodied memory and proprioceptive awareness to make digital functions more intuitive and memorable [1].

A third strand addresses ambiguity in cluttered or occluded environments, where conventional ray-based pointing often leads to errors and frustration. To resolve this, I developed a hybrid technique that combines expressive Bézier trajectories with proprioceptive on-body disambiguation. Users bend curves through finger flexion to reach around obstacles, while nearby candidate targets are projected

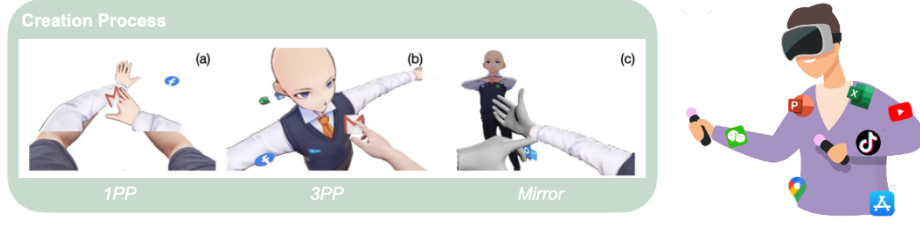


Figure 2: On-body menus [1] in VR created from first-person, third-person, and mirrored perspectives. These perspectives shape how users place and recall functions, while consistent patterns of icon placement reveal strong associations between application categories and bodily landmarks.

onto the forearm for tactile confirmation. This combination merges mid-air flexibility with embodied verification: the curve expands the space of reachable selections, and the on-body projection anchors disambiguation in a proprioceptive channel. Empirical evaluations demonstrate that this method reduces errors and subjective workload compared to ray casting, while qualitative feedback emphasises the confidence and precision gained through proprioceptive grounding [3, 4].

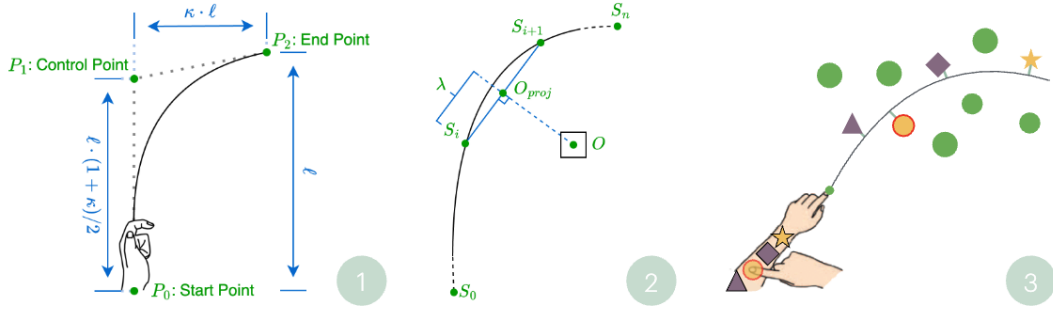


Figure 3: A hybrid selection technique that combines curved Bézier trajectories with on-body disambiguation [3, 4]. Users bend a virtual path through finger flexion to reach occluded targets, while nearby candidates are projected onto the forearm for tactile confirmation.

The final strand refines bodily interaction into subtle, efficient microgestures for everyday productivity. I introduced microGEXT, a lightweight framework for VR text editing that uses small, ergonomic finger and wrist movements detected by built-in headset cameras. Commands such as copy, paste, and delete can be executed without fatigue, overcoming the limitations of large mid-air gestures that cause discomfort during prolonged use. Across three studies, microGEXT demonstrates faster performance in many editing tasks, lower workload, and higher user preference compared to conventional gesture–menu systems. Participants reported that once learned, microgestures enabled smoother task flows and significantly reduced physical strain [2]. These findings show that bodily interaction need not be expansive or dramatic to be effective; when deliberately minimised, it can deliver precision, comfort, and sustainability for long-term creative and professional use.

Together, these four directions form a coherent research programme that reconceptualises the body from a representational proxy into an intelligent interface. Methodologically, the work combines iterative prototyping with rigorous empirical evaluation, drawing on both quantitative measures of performance and qualitative insights into user experience. Conceptually, it demonstrates how proprioceptive awareness, spatial memory, and bodily schemas can be mobilised as design resources. Practically, it offers a repertoire of techniques that advance XR beyond entertainment toward productivity,



Figure 4: Comparing microGEXT [2], a lightweight microgesture-based framework for VR text editing, with a conventional gesture-and-menu baseline. Subtle hand movements enable commands like copy, paste, or delete with reduced fatigue and smoother task flow.

collaboration, and sustainable everyday use.

FUTURE RESEARCH

Building on this foundation, my research agenda is to investigate how **AI-driven context awareness** can enable adaptive XR interfaces that sense spatial, social, and task context. The overarching vision is to design immersive systems that are embodied yet also adaptive and intelligent, capable of responding fluidly to environments, social situations, and evolving user goals. If the body is the foundation of interaction, then AI can serve as the adaptive layer that tunes interaction to context, ensuring that immersive systems remain sustainable and seamlessly embedded into everyday life. I will build on my ongoing collaborations with Sony, Google, Meta, the University of Cambridge, the University of Tokyo, the University of Konstanz, Monash, and UCL, alongside colleagues at CUHK-Shenzhen (i.e., School of AI, School of Data Science, and School of Science and Engineering), to pursue this vision in a global and interdisciplinary context.

Spatially Adaptive Interfaces. A first strand of this agenda investigates how immersive systems can adapt to the geometry and affordances of physical environments. AI-driven spatial adaptation pipelines can integrate environment sensing with embodied interaction, enabling interfaces that dynamically reconfigure their scale, modality, or trajectories based on context. This direction will allow XR systems to move beyond static layouts toward fluid, environment-aware companions. My forthcoming internship at Sony CSL in Kyoto will provide an initial step, where I will prototype adaptive MR systems that combine embodied interaction with AI-enabled environmental intelligence.

AI Agents for Social XR. A second strand explores how AI agents can actively participate in or mediate social interaction in XR. Embodied agents—autonomous or semi-autonomous entities capable of perceiving, reasoning, and acting—could serve as partners, mediators, or orchestrators in collaborative contexts. Unlike static avatars, these agents would be adaptive, drawing on multimodal signals (gaze, gestures, speech, proxemics) to model group state and intervene appropriately. This vision moves beyond social acceptability toward socially intelligent XR environments, where agents enable richer collaboration, new forms of shared embodiment, and more equitable distribution of control.

Productivity and Well-being. A third strand considers how adaptive XR systems can sustain productivity and well-being over time. My prior work on microgestures demonstrated the feasibility

of lightweight and fatigue-free input, but real-world scenarios demand systems that adapt dynamically to shifting tasks. I aim to develop task-aware MR systems that can privilege different modalities (e.g., microgestures for focused editing, voice for brainstorming), or introduce restorative interventions (e.g., gamified micro-breaks when fatigue is detected). By uniting natural language models with physiological sensing, these systems can support efficiency while maintaining long-term human well-being.

Infrastructure and Conclusion. My approach combines iterative prototyping with AI-driven adaptation pipelines, reinforcement learning for interface optimisation, multimodal signal fusion for context inference, embodied agent design for social XR, and longitudinal deployment studies for ecological validation. To realise this vision, access to state-of-the-art XR infrastructure is essential. This includes the latest VR and MR headsets with integrated eye, hand, and body tracking; external motion capture systems for high-fidelity ground truth data; and high-performance workstations equipped with next-generation graphics cards (e.g., NVIDIA RTX 4090/5090) to support real-time rendering, AI inference, and multimodal data processing. A flexible experimental laboratory space is also required to support user studies, equipped with reconfigurable physical setups, haptic devices, and prototyping toolkits (e.g., 3D printers, microcontrollers, sensors) for rapid iteration.

My broader vision is to establish new theoretical foundations for embodied intelligence, practical design principles for adaptive XR interfaces, and international collaborations that ensure this research is globally connected, ambitious, and human-centred.

References

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